Extended E2 (E2X): results

Built by `scripts/e2x/make_report.py` from the files listed at the end. Preregistration: tag `e2x-prereg` (`experiments/E2X.md` §5). DESCRIPTIVE parts are labelled; calibration-clip numbers are not results.

> **ANALYSIS PAPER (no method claim): E3 is NOT CALIBRATED for MapAnything, so the photo witness cannot be made safe on correct hints under the fixed rule. E3 is reported as a negative result.**

Decisions (rules fixed in advance)

E2X continue / stop rule (E2X.md §5; primary calibration)

Signal
(i-a) real whole-room pose hints HURT for ≥ 1 model (X-B, dev_pose, rotation or shape)
(i-b) E3 removes more damage than the best of R0/R2/R3 at ≥ its benefit kept (MapAnything, cal clips; R1 'drop all poses' is not in the
(ii) E3 calibrated for MapAnything

> **ANALYSIS PAPER (no method claim): E3 is NOT CALIBRATED for MapAnything, so the photo witness cannot be made safe on correct hints under the fixed rule. E3 is reported as a negative result.**

E3 method evaluation (E3.md; thresholds from the accurate-ground-truth calibration)

> **E3 method claim holds: False (SAFE: True). Both readings are reported: the primary rule above and this evaluation.**

Inside-the-network direction: Shubh's G1 oracle dial gave no reading in session A (its own γ = 1 check failed). Session A2 measured the same dial (M1, section 6).

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1. The session

```
NVIDIA A100-SXM4-80GB, 580.178.04, 81920 MiB, 0 MiB
cpus 12
Mem:             167           5         112           0          50         161
/dev/root       991G   66G  926G   7% /
torch wheels cu130
our GPU budget 32 GB
```

First and last queue lines:

```
2026-11-07 18:38:51 start e2x session A, budget 32 GB of 80 GB, free now 15 GB, tree
ab5bc73d5a844da74539d390883b6273ad6b3482, scripts 1c0835f8c12bac528b08c2d16a6de124d1f0fb6c
2026-11-08 08:48:59 PART_PX_DONE
```

AI depth check against E2-E's record: `AI depth on the E2-E clips: 0/49 bit-identical to E2-E; same model revision: True`

Runs per job

run	rows	ok	failed rows	failures.jsonl lines	out-of-memory resets
e2x_e2e_px_dev	168	168	0	0	0
e2x_e2e_px_eval	861	861	0	0	0
e2x_e3cal_real_ma_cal	2050	2050	0	0	0
e2x_e3cal_real_ma_roomdev	287	287	0	0	0
e2x_e3cal_real_px_cal	2050	2050	0	0	0
e2x_e3cal_synth_ma_cal	16000	16000	0	0	27
e2x_e3cal_synth_ma_eth3d	960	960	0	0	0
e2x_e3cal_synth_ma_tum	209	209	0	0	0
e2x_e3cal_synth_ov_eth3d	960	960	0	0	0
e2x_e3cal_synth_px_cal	16000	16000	0	0	0
e2x_e3cal_synth_px_eth3d	960	960	0	0	0
e2x_e3cal_synth_px_tum	209	209	0	0	0
e2x_pi3x_checks	129	129	0	0	0
e2x_pi3x_eth3d_dev	1386	1386	0	0	0
e2x_pi3x_eth3d_eval	11088	11088	0	0	0
e2x_room_ma_dev	147	147	0	0	0
e2x_room_ma_eval	756	756	0	0	0
e2x_room_ov_dev	133	133	0	0	0
e2x_room_ov_eval	684	684	0	0	0
e2x_room_px_dev	147	147	0	0	0
e2x_room_px_eval	756	756	0	0	0

Dev gates and job markers

```
2026-11-07 19:06:08 GATE_MA had errors: see /home/dawar/e2x_logs/gate.txt (queue continues)
GATE e2e results/e2x_room_ov_dev: PASS
GATE ov_e2d results/e2x_pi3x_eth3d_dev: PASS
GATE e2e results/e2x_room_ma_dev: PASS
GATE e2e results/e2x_e2e_px_dev: PASS
GATE e2e results/e2x_room_px_dev: PASS
2026-11-08 01:27:07   !! 27 runs reset after CUDA out-of-memory; all processes ok: 1 (logs:
results/e2x_e3cal_synth_ma_cal/_shards/*/log.txt). Retry in 120 s
2026-11-08 01:40:35   Pi3X peak measured: 9 GB + headroom -> 10 GB per process
2026-11-08 02:38:20 ROOM_OV_EVAL_DONE
2026-11-08 03:22:04 ROOM_MA_EVAL_DONE
```

```
2026-11-08 03:26:37 PX_ETH3D_EVAL_DONE
2026-11-08 03:59:25 PX_E2E_EVAL_DONE
2026-11-08 04:31:03 ROOM_PX_EVAL_DONE
2026-11-08 08:48:59 PART_PX_DONE
```

---

2. X-A: Pi3X, the third model (ETH3D)

Preregistered D-H test (E2-D verbatim, Holm within the model): **5** of 56 tests harmful, families: depth_one_scale, drift, outlier_frac, outlier_rot.

condition	metric	x_mean	floor_mean	delta	ci_lo	ci_hi	n_worse	p_holm
depth_one_scale_factor=1.5	shape	0.071	0.021	0.051	0.041	0.060	10	0.0002
outlier_rot_rot_deg=90	rotation	8.373	0.491	7.882	6.002	9.824	10	0.0018
outlier_rot_rot_deg=30	rotation	1.337	0.491	0.846	0.579	1.103	10	0.0114
drift_level=3	rotation	1.910	0.491	1.419	0.976	1.883	10	0.0133
outlier_frac_frac=0.4	rotation	2.007	0.491	1.516	1.052	2.058	10	0.0192

Behaviour (P1–P5, P10, P11)

test	question	mean	ci_lo	ci_hi	n_scenes	p_holm	finding
P1_rot_override	C1	0.082	0.027	0.145	10	0.1559	False
P2_trans_override	C1	0.446	0.347	0.539	10	0.0001	False
P3_conf_flags_bad_view	C2	-0.015	-0.106	0.081	10	0.7823	False
P4_bad_view_own_shape	C3	0.000	0.000	0.001	10	0.1803	False
P5_good_views_shape	C3	0.000	0.000	0.000	10	0.1803	False
P10_shared_vs_indep_conf_5deg	C6	0.004	0.002	0.007	10	0.1065	False
P11_drift_vs_jitter_conf_level2	C6	0.001	0.000	0.002	10	0.1803	False

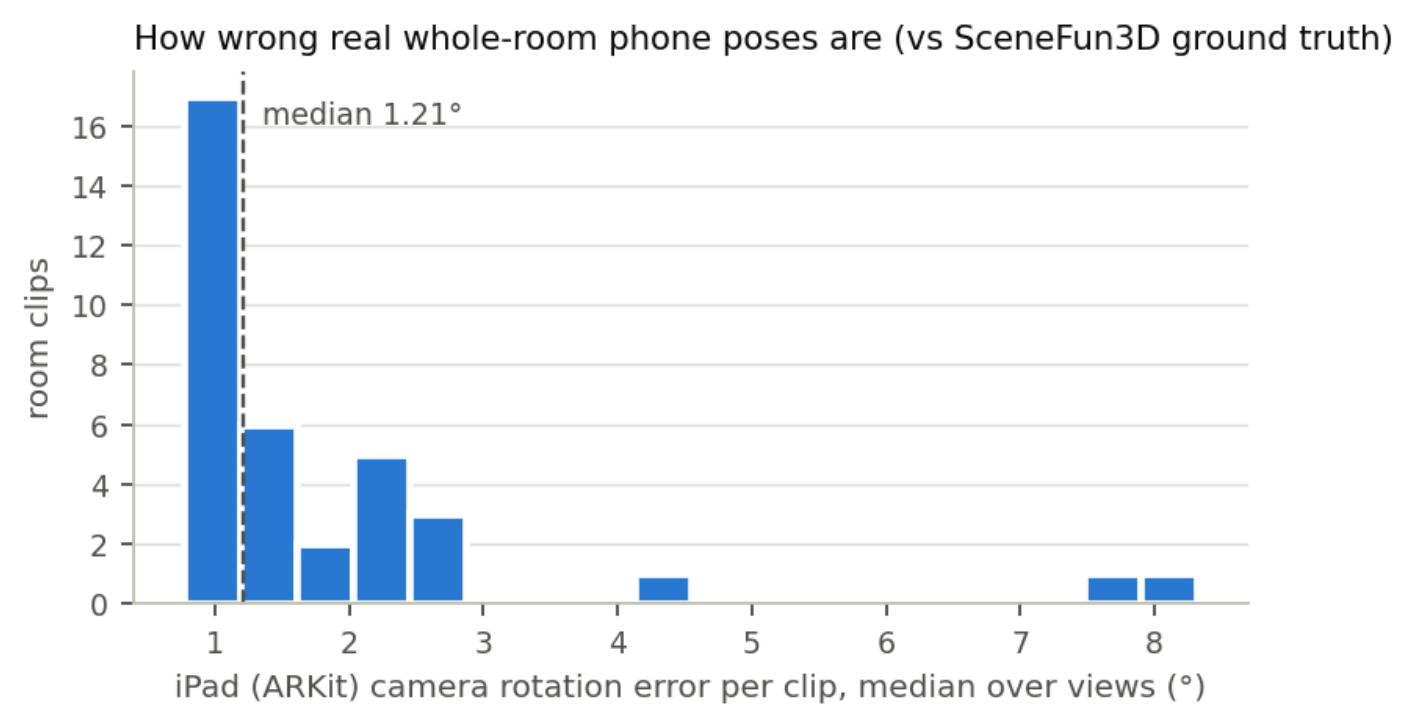
Per family: MapAnything vs Pi3X

family	class
anchor_error	neither
depth_one_scale	same
depth_scale	MapAnything only
drift	same
noise_rot	neither
outlier_frac	same
outlier_rot	same
scale_all	MapAnything only
shared_heading_bias	neither

Behaviour side by side

test	ma_mean	ma_finding	ov_mean	ov_finding	class
P1_rot_override	0.028	False	0.082	False	neither
P2_trans_override	0.112	False	0.446	False	neither
P3_conf_flags_bad_view	0.431	True	-0.015	False	MapAnything only
P4_bad_view_own_shape	0.006	False	0.000	False	neither
P5_good_views_shape	0.000	False	0.000	False	neither
P10_shared_vs_indep_conf_5deg	0.252	True	0.004	False	MapAnything only
P11_drift_vs_jitter_conf_level2	0.280	True	0.001	False	MapAnything only

3. X-B: real hints over whole rooms



36 room clips: median rotation error 1.21° (clip medians), share of views ≥ 5° 0.06, depth scale ratio LiDAR/GT median 0.995 (`results/e2x_agg/xb_room_real_errors.csv`).

MapAnything: each real hint set vs images only (room eval clips)

condition	metric	delta	ci_lo	ci_hi	n_scenes	p_holm	verdict
dev_K	shape	-0.011	-0.022	0.000	15	0.6777	no clear effect
dev_K	rotation	0.213	-0.794	1.490	15	1.0000	no clear effect
dev_pose	shape	-0.162	-0.292	-0.072	15	0.1536	no clear effect
dev_pose	rotation	-15.210	-24.785	-6.760	15	0.0953	no clear effect
dev_pose	real_size	-0.449	-0.619	-0.323	15	0.0011	HELPS
dev_depth	shape	-0.072	-0.165	-0.014	15	0.7267	no clear effect
dev_depth	rotation	-4.018	-9.456	-0.021	15	0.7267	no clear effect
dev_depth	real_size	-0.369	-0.506	-0.266	15	0.0011	HELPS
dev_all	shape	-0.186	-0.322	-0.089	15	0.1166	no clear effect
dev_all	rotation	-15.220	-24.784	-6.776	15	0.0953	no clear effect
dev_all	real_size	-0.482	-0.658	-0.349	15	0.0009	HELPS
ai_depth	shape	0.053	-0.046	0.132	15	1.0000	no clear effect
ai_depth	rotation	-4.236	-9.116	-0.216	15	0.6924	no clear effect
ai_depth	real_size	-0.039	-0.178	0.095	15	1.0000	no clear effect
dev_pose_ai_depth	shape	-0.014	-0.138	0.078	15	1.0000	no clear effect
dev_pose_ai_depth	rotation	-15.221	-24.780	-6.776	15	0.0953	no clear effect
dev_pose_ai_depth	real_size	-0.231	-0.399	-0.112	15	0.1166	no clear effect

Remedies on the real hints (Δ vs no remedy):

condition	remedy	metric	delta_vs_no_remedy	ci_lo	ci_hi	share_views_dropped
dev_pose	filter_rgbcheck	shape	0.137	0.045	0.272	0.321
dev_pose	filter_rgbcheck	rotation	11.915	4.156	20.945	0.321
dev_pose	drop_lowconf	shape	0.006	0.001	0.013	0.037
dev_pose	drop_lowconf	rotation	0.866	0.030	2.454	0.037
dev_all	filter_rgbcheck	shape	0.096	0.042	0.160	0.321
dev_all	filter_rgbcheck	rotation	9.455	3.760	16.122	0.321
dev_all	drop_lowconf	shape	0.007	0.002	0.012	0.074
dev_all	drop_lowconf	rotation	1.850	0.346	3.770	0.074
dev_pose_ai_depth	filter_rgbcheck	shape	0.053	0.017	0.096	0.321
dev_pose_ai_depth	filter_rgbcheck	rotation	7.915	3.177	13.494	0.321
dev_pose_ai_depth	drop_lowconf	shape	-0.007	-0.011	-0.003	0.086
dev_pose_ai_depth	drop_lowconf	rotation	1.760	0.410	3.590	0.086

OmniVGGT: each real hint set vs images only (room eval clips)

condition	metric	delta	ci_lo	ci_hi	n_scenes	p_holm	verdict
dev_pose	shape	-0.130	-0.255	-0.039	15	0.2606	no clear effect
dev_pose	rotation	-11.757	-20.282	-4.504	15	0.1246	no clear effect
dev_depth	shape	0.028	-0.004	0.066	15	0.4450	no clear effect
dev_depth	rotation	0.695	-2.052	2.897	15	1.0000	no clear effect
dev_all	shape	-0.122	-0.243	-0.037	15	0.2606	no clear effect
dev_all	rotation	-11.749	-20.287	-4.521	15	0.1246	no clear effect
ai_depth	shape	0.128	0.089	0.178	15	0.0010	HURTS
ai_depth	rotation	2.935	0.155	5.965	15	0.2923	no clear effect
dev_pose_ai_depth	shape	-0.038	-0.157	0.046	15	1.0000	no clear effect
dev_pose_ai_depth	rotation	-11.502	-20.048	-4.264	15	0.1246	no clear effect

Remedies on the real hints (Δ vs no remedy):

condition	remedy	metric	delta_vs_no_remedy	ci_lo	ci_hi	share_views_dropped
dev_pose	filter_rgbcheck	shape	0.097	0.018	0.196	0.225
dev_pose	filter_rgbcheck	rotation	9.243	3.082	16.669	0.225
dev_pose	drop_lowconf	shape	0.000	-0.000	0.000	0.009
dev_pose	drop_lowconf	rotation	0.017	0.000	0.036	0.009
dev_all	filter_rgbcheck	shape	0.086	0.021	0.171	0.225
dev_all	filter_rgbcheck	rotation	10.443	4.125	18.110	0.225
dev_all	drop_lowconf	shape	0.000	0.000	0.000	0.000
dev_all	drop_lowconf	rotation	0.000	0.000	0.000	0.000
dev_pose_ai_depth	filter_rgbcheck	shape	0.087	0.023	0.170	0.225
dev_pose_ai_depth	filter_rgbcheck	rotation	10.996	4.868	18.299	0.225
dev_pose_ai_depth	drop_lowconf	shape	-0.000	-0.001	0.000	0.003
dev_pose_ai_depth	drop_lowconf	rotation	0.016	0.000	0.049	0.003

Pi3X: each real hint set vs images only (room eval clips)

condition	metric	delta	ci_lo	ci_hi	n_scenes	p_holm	verdict
dev_K	shape	0.014	-0.008	0.040	15	0.8173	no clear effect
dev_K	rotation	0.504	-0.106	1.378	15	0.8173	no clear effect
dev_pose	shape	-0.162	-0.289	-0.056	15	0.1554	no clear effect
dev_pose	rotation	-6.375	-10.917	-2.559	15	0.1260	no clear effect
dev_depth	shape	0.073	-0.022	0.187	15	0.8173	no clear effect
dev_depth	rotation	2.580	-0.325	6.029	15	0.7634	no clear effect
dev_all	shape	-0.165	-0.293	-0.057	15	0.1554	no clear effect
dev_all	rotation	-6.515	-11.023	-2.729	15	0.1227	no clear effect
ai_depth	shape	0.200	0.109	0.307	15	0.0230	HURTS
ai_depth	rotation	5.547	2.184	9.386	15	0.1260	no clear effect
dev_pose_ai_depth	shape	-0.000	-0.123	0.103	15	0.9989	no clear effect
dev_pose_ai_depth	rotation	-5.796	-10.365	-1.981	15	0.1554	no clear effect

Remedies on the real hints (Δ vs no remedy):

condition	remedy	metric	delta_vs_no_remedy	ci_lo	ci_hi	share_views_dropped
dev_pose	filter_rgbcheck	shape	0.183	0.052	0.341	0.290
dev_pose	filter_rgbcheck	rotation	6.783	2.467	11.719	0.290
dev_pose	drop_lowconf	shape	0.000	0.000	0.000	0.000
dev_pose	drop_lowconf	rotation	0.000	0.000	0.000	0.000
dev_all	filter_rgbcheck	shape	0.225	0.034	0.457	0.290
dev_all	filter_rgbcheck	rotation	8.018	2.338	14.897	0.290
dev_all	drop_lowconf	shape	-0.000	-0.000	0.000	0.003
dev_all	drop_lowconf	rotation	-0.000	-0.001	0.000	0.003
dev_pose_ai_depth	filter_rgbcheck	shape	0.205	0.037	0.413	0.290
dev_pose_ai_depth	filter_rgbcheck	rotation	9.704	3.682	16.628	0.290
dev_pose_ai_depth	drop_lowconf	shape	0.000	0.000	0.000	0.000
dev_pose_ai_depth	drop_lowconf	rotation	0.000	0.000	0.000	0.000

Pi3X on the E2-E short clips

condition	metric	delta	ci_lo	ci_hi	n_scenes	p_holm	verdict
dev_K	shape	-0.010	-0.016	-0.006	15	0.0126	HELPS
dev_K	rotation	-0.231	-0.305	-0.163	15	0.0002	clear but below the practical threshold
dev_pose	shape	-0.012	-0.020	-0.006	15	0.0130	HELPS
dev_pose	rotation	-0.297	-0.386	-0.216	15	0.0001	HELPS
dev_depth	shape	-0.004	-0.013	0.006	15	0.4037	no clear effect
dev_depth	rotation	-0.159	-0.277	-0.019	15	0.0689	no clear effect
dev_all	shape	-0.010	-0.018	-0.004	15	0.0463	HELPS
dev_all	rotation	-0.300	-0.391	-0.218	15	0.0001	HELPS
ai_depth	shape	0.115	0.086	0.146	15	0.0000	HURTS
ai_depth	rotation	1.136	0.935	1.343	15	0.0000	HURTS
dev_pose_ai_depth	shape	0.106	0.085	0.128	15	0.0000	HURTS
dev_pose_ai_depth	rotation	0.260	0.123	0.393	15	0.0130	HURTS

4. E3 calibration (calibration and dev clips only; nothing here is a result)

Rule fixed before the session (E2X.md §5): the smallest `e3_z` whose false-action rate on ground-truth hints is ≤ 5% for any group action and ≤ 2% of views for single-view drops; the same for the depth cut-offs; otherwise NOT CALIBRATED, never loosened.

MapAnything

Chosen: `{"e3_z": "NOT CALIBRATED", "e3_tau_depth_shared": 0.2, "e3_tau_depth_view": 0.25, "n_gt_pose_decisions": 269, "n_gt_depth_decisions": 110}`

part	setting	n	group_action_rate	view_drop_rate	meets_rule
pose	e3_z=2.0	269	0.599	0.118	False
pose	e3_z=2.5	269	0.461	0.061	False
pose	e3_z=3.0	269	0.398	0.044	False
pose	e3_z=4.0	269	0.249	0.038	False
pose	e3_z=5.0	269	0.138	0.033	False
depth_shared	e3_tau_depth_shared=0.05	110	0.518	–	False
depth_shared	e3_tau_depth_shared=0.1	110	0.255	–	False
depth_shared	e3_tau_depth_shared=0.15	110	0.100	–	False
depth_shared	e3_tau_depth_shared=0.2	110	0.036	–	True
depth_view	e3_tau_depth_view=0.1	110	–	0.089	False
depth_view	e3_tau_depth_view=0.15	110	–	0.046	False
depth_view	e3_tau_depth_view=0.25	110	–	0.019	True

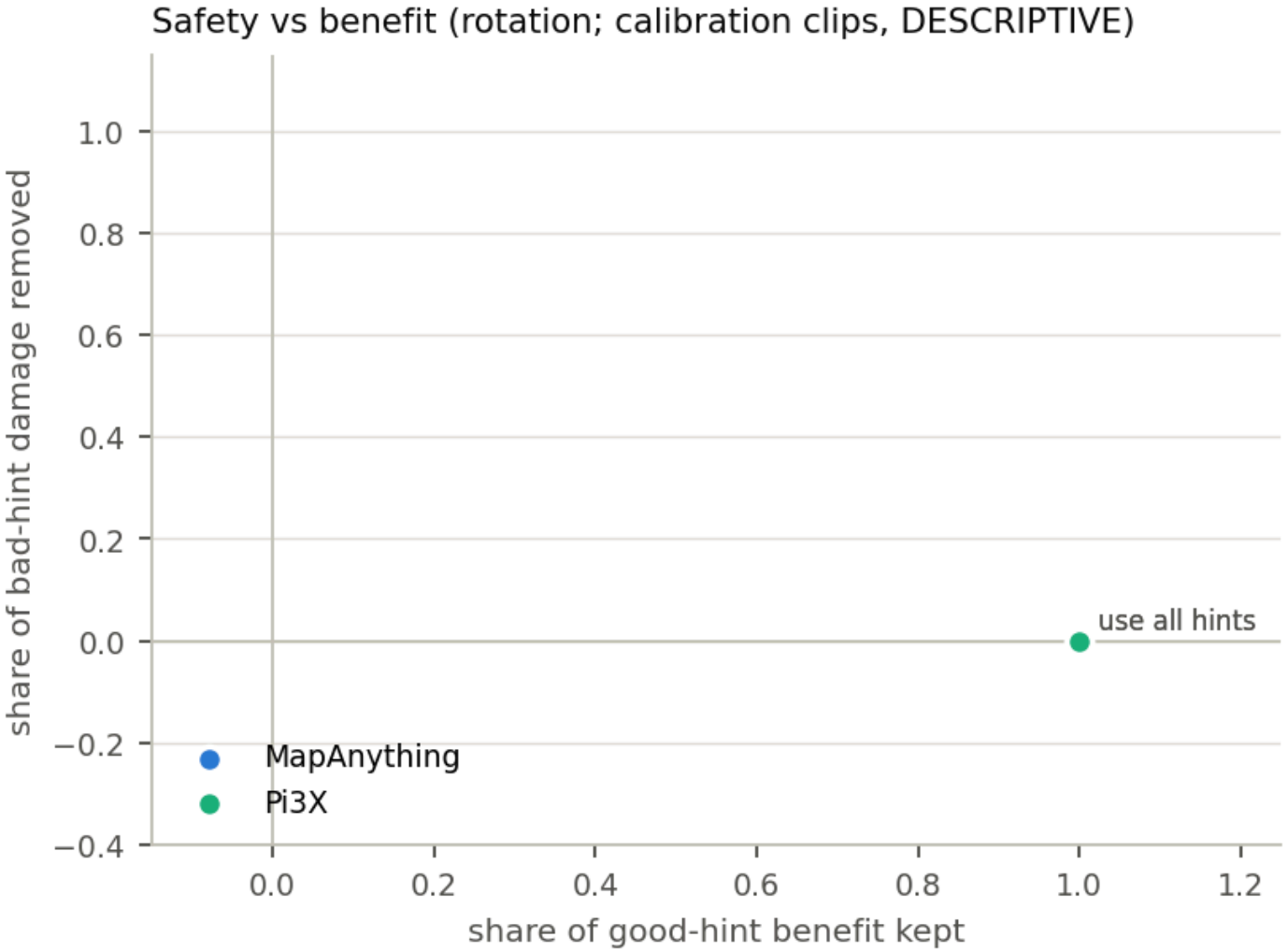
Pi3X

Chosen: {"e3_z": "NOT CALIBRATED", "e3_tau_depth_shared": 0.15, "e3_tau_depth_view": 0.15, "n_gt_pose_decisions": 206, "n_gt_depth_decisions": 103}

part	setting	n	group_action_rate	view_drop_rate	meets_rule
pose	e3_z=2.0	206	0.786	0.157	False
pose	e3_z=2.5	206	0.757	0.119	False
pose	e3_z=3.0	206	0.680	0.104	False
pose	e3_z=4.0	206	0.485	0.091	False
pose	e3_z=5.0	206	0.388	0.076	False
depth_shared	e3_tau_depth_shared=0.05	103	0.272	–	False
depth_shared	e3_tau_depth_shared=0.1	103	0.087	–	False
depth_shared	e3_tau_depth_shared=0.15	103	0.039	–	True
depth_shared	e3_tau_depth_shared=0.2	103	0.039	–	True
depth_view	e3_tau_depth_view=0.1	103	–	0.036	False
depth_view	e3_tau_depth_view=0.15	103	–	0.017	True
depth_view	e3_tau_depth_view=0.25	103	–	0.007	True

OmniVGGT

Not available: results/e2x_e3cal_agg/omnivggt/chosen.json (input file missing; nothing is estimated).


Safety vs benefit (rotation; calibration clips, DESCRIPTIVE)
share of bad-hint damage removed
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
use all hints
MapAnything
Pi3X
0.0
0.2
0.4
0.6
0.8
1.0
1.2
share of good-hint benefit kept

model	metric	remedy	benefit_kept	damage_removed	n_bad_conditions
mapanything	rre_deg	drop_lowconf	0.938	-1.471	21
mapanything	rre_deg	filter_rgbcheck	0.321	-41.025	21
mapanything	rre_deg	none	1.000	0.000	21
mapanything	rre_deg	witness	0.550	-23.558	21
mapanything	rre_deg	witness_drop	0.358	-33.489	21
mapanything	rre_deg	R0 images only	0.000	-49.565	–
mapanything	pts_relerr_scale	drop_lowconf	0.964	-4.271	21
mapanything	pts_relerr_scale	filter_rgbcheck	0.371	-70.622	21
mapanything	pts_relerr_scale	none	1.000	0.000	21
mapanything	pts_relerr_scale	witness	0.623	-38.318	21
mapanything	pts_relerr_scale	witness_drop	0.353	-63.249	21
mapanything	pts_relerr_scale	R0 images only	0.000	-99.402	–
pi3x	rre_deg	drop_lowconf	0.987	-1.279	21
pi3x	rre_deg	filter_rgbcheck	0.274	-37.356	21
pi3x	rre_deg	none	1.000	0.000	21
pi3x	rre_deg	witness	0.237	-34.644	21
pi3x	rre_deg	witness_drop	0.048	-39.734	21
pi3x	rre_deg	R0 images only	0.000	-41.074	–
pi3x	pts_relerr_scale	drop_lowconf	0.984	-5.715	21
pi3x	pts_relerr_scale	filter_rgbcheck	0.399	-134.425	21
pi3x	pts_relerr_scale	none	1.000	0.000	21
pi3x	pts_relerr_scale	witness	0.456	-135.224	21
pi3x	pts_relerr_scale	witness_drop	0.211	-165.511	21
pi3x	pts_relerr_scale	R0 images only	0.000	-177.009	–

5b. E3 calibration on accurate ground truth (E3-cal amendment 1, SECONDARY)

Written before any TUM run (tag `e3cal-amend1`): the same rule on ETH3D dev (laser scan) and TUM RGB-D fr3 (motion capture) clips; TUM clips failing the E0 ground-truth check were excluded first (`results/e2x_tum_clip_check/excluded.csv`). The primary verdict (section 4) stands.

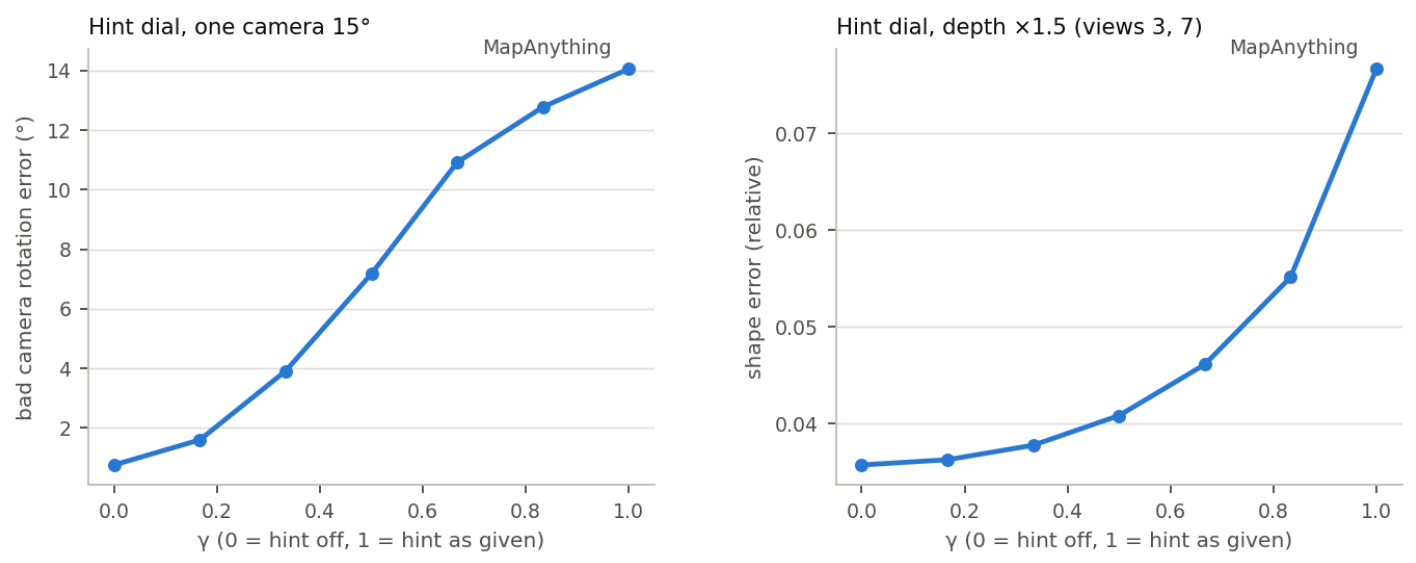
model	calibration	e3_z	tau_depth_shared	tau_depth_view	n_gt_pose_decisions	n_gt_depth_de
MapAnything	primary	NOT CALIBRATED	0.2	0.25	269	
MapAnything	secondary	5.0	0.1	0.1	44	
Pi3X	primary	NOT CALIBRATED	0.15	0.15	206	
Pi3X	secondary	4.0	0.05	0.1	44	

▶ MapAnything: secondary false-action rates

▶ Pi3X: secondary false-action rates

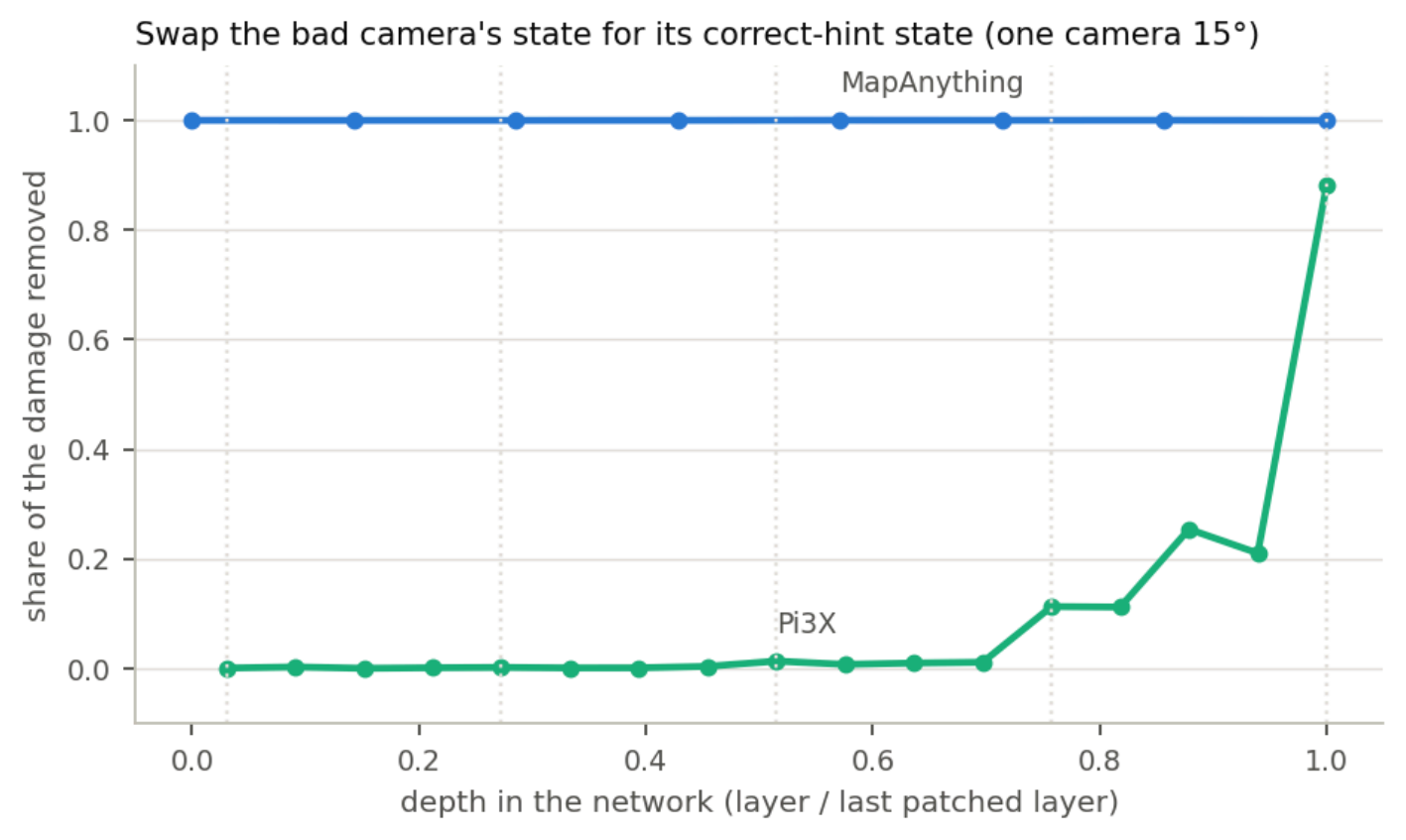
6. Inside the network (session A2, DESCRIPTIVE; ETH3D eval, 24 clips, 10 scenes)

Frozen at `e2x-a2-prereg` (amendments 1–2: plumbing fixes before any eval run). Pi3X injects the camera hint again after blocks 1, 9, 17, 25, 33, so a swapped state can be overwritten later (stated before the run); the dotted lines mark those blocks.



M1 hint dial

model	case	r2_mean	r2_ci_lo	r2_ci_hi	n_scenes_linear	switch_index_mean	n_scenes_switch
mapanything	depth_x1.5	0.678	0.534	0.785	0	-0.013	0
mapanything	drift_level2	0.595	0.397	0.776	1	0.143	1
mapanything	one_cam_15	0.944	0.914	0.971	7	0.063	0
mapanything	shared_heading_5	0.634	0.461	0.785	0	0.154	1



M3 point of no return (first layer whose swap removes < 50% of the damage)

model	case	source	point_of_no_return_layer
mapanything	depth_x1.5	clean	8.0
mapanything	depth_x1.5	drop	8.0
mapanything	drift_level2	clean	none
mapanything	drift_level2	drop	8.0
mapanything	one_cam_15	clean	none
mapanything	one_cam_15	drop	none
mapanything	shared_heading_5	clean	none
mapanything	shared_heading_5	drop	8.0
pi3x	depth_x1.5	clean	1.0
pi3x	depth_x1.5	drop	1.0
pi3x	drift_level2	clean	1.0
pi3x	drift_level2	drop	1.0
pi3x	one_cam_15	clean	1.0
pi3x	one_cam_15	drop	1.0
pi3x	shared_heading_5	clean	1.0
pi3x	shared_heading_5	drop	1.0

▶ M3 share removed per layer (source: correct-hint run)

▶ M5: swapping the OTHER cameras' states

▶ M2/M4: size of the bad − correct state difference and its cosine with the first layer

M8: two hints disagree (output scale vs ground truth)

model	condition	pose_hint_scale	depth_hint_scale	output_scale_vs_gt	ci_lo	ci_hi	shape_err
mapanything	clean_pose_depth2	–	–	1.025	1.010	1.041	0.031
mapanything	m8_pose0.67_depth0.67	0.67	0.67	0.713	0.680	0.740	0.059
mapanything	m8_pose0.67_depth1.0	0.67	1.00	0.824	0.771	0.873	0.106
mapanything	m8_pose0.67_depth1.5	0.67	1.50	1.058	0.971	1.141	0.191
mapanything	m8_pose1.0_depth0.67	1.00	0.67	0.943	0.918	0.963	0.066
mapanything	m8_pose1.0_depth1.0	1.00	1.00	1.025	1.010	1.041	0.031
mapanything	m8_pose1.0_depth1.5	1.00	1.50	1.157	1.126	1.190	0.077
mapanything	m8_pose1.5_depth0.67	1.50	0.67	1.121	1.039	1.208	0.209
mapanything	m8_pose1.5_depth1.0	1.50	1.00	1.319	1.267	1.367	0.086
mapanything	m8_pose1.5_depth1.5	1.50	1.50	1.450	1.426	1.474	0.034
mapanything	rgb_only	–	–	1.179	1.069	1.309	0.164
pi3x	clean_pose_depth2	–	–	0.887	0.744	1.017	0.013
pi3x	m8_pose0.67_depth0.67	0.67	0.67	0.887	0.744	1.017	0.013
pi3x	m8_pose0.67_depth1.0	0.67	1.00	0.879	0.736	1.010	0.018
pi3x	m8_pose0.67_depth1.5	0.67	1.50	0.885	0.741	1.016	0.085
pi3x	m8_pose1.0_depth0.67	1.00	0.67	0.909	0.766	1.041	0.022
pi3x	m8_pose1.0_depth1.0	1.00	1.00	0.887	0.744	1.017	0.013
pi3x	m8_pose1.0_depth1.5	1.00	1.50	0.879	0.736	1.009	0.018
pi3x	m8_pose1.5_depth0.67	1.50	0.67	0.895	0.753	1.027	0.206
pi3x	m8_pose1.5_depth1.0	1.50	1.00	0.910	0.766	1.041	0.022
pi3x	m8_pose1.5_depth1.5	1.50	1.50	0.887	0.744	1.017	0.013
pi3x	rgb_only	–	–	0.836	0.700	0.959	0.094

7. E3 method evaluation, MapAnything (experiments/E3.md, eval splits)

E3 decision (fixed before the run): method claim holds = False; SAFE = True; HELP on agreeing / AI-depth: c6_shared_rot_rot_deg=5, dev_pose_ai_depth, drift_level=2, drift_level=3, shared_heading_bias_rot_deg=10, shared_heading_bias_rot_deg=5; BEST on agreeing / AI-depth: none.

Caution: SAFE passes under the strict rule, but 13 SAFE tests show harm whose 95% CI excludes 0 and whose estimate exceeds the practical threshold, without surviving Holm: real_room gt_pose shape: +0.059 (CI 0.011 to 0.136); real_room gt_pose rotation: +4.032 (CI 0.935 to 7.759); real_room gt_pose real_size: +0.063 (CI 0.012 to 0.144); real_room gt_all shape: +0.070 (CI 0.017 to 0.153); real_room gt_all rotation: +4.073 (CI 0.969 to 7.803); real_room gt_all real_size: +0.073 (CI 0.017 to 0.157); real_room dev_pose shape: +0.055 (CI 0.007 to 0.130); real_room dev_pose rotation: +3.833 (CI 0.767 to 7.515); real_room dev_pose real_size: +0.060 (CI 0.009 to 0.141); real_room dev_all shape: +0.061 (CI 0.009 to 0.142); real_room dev_all rotation: +3.212 (CI

0.411 to 6.918); real_room dev_all real_size: +0.062 (CI 0.009 to 0.144); real_e2e gt_all rotation: +0.984 (CI 0.002 to 2.775). E3 is therefore not shown to be safe on these data.

SAFE

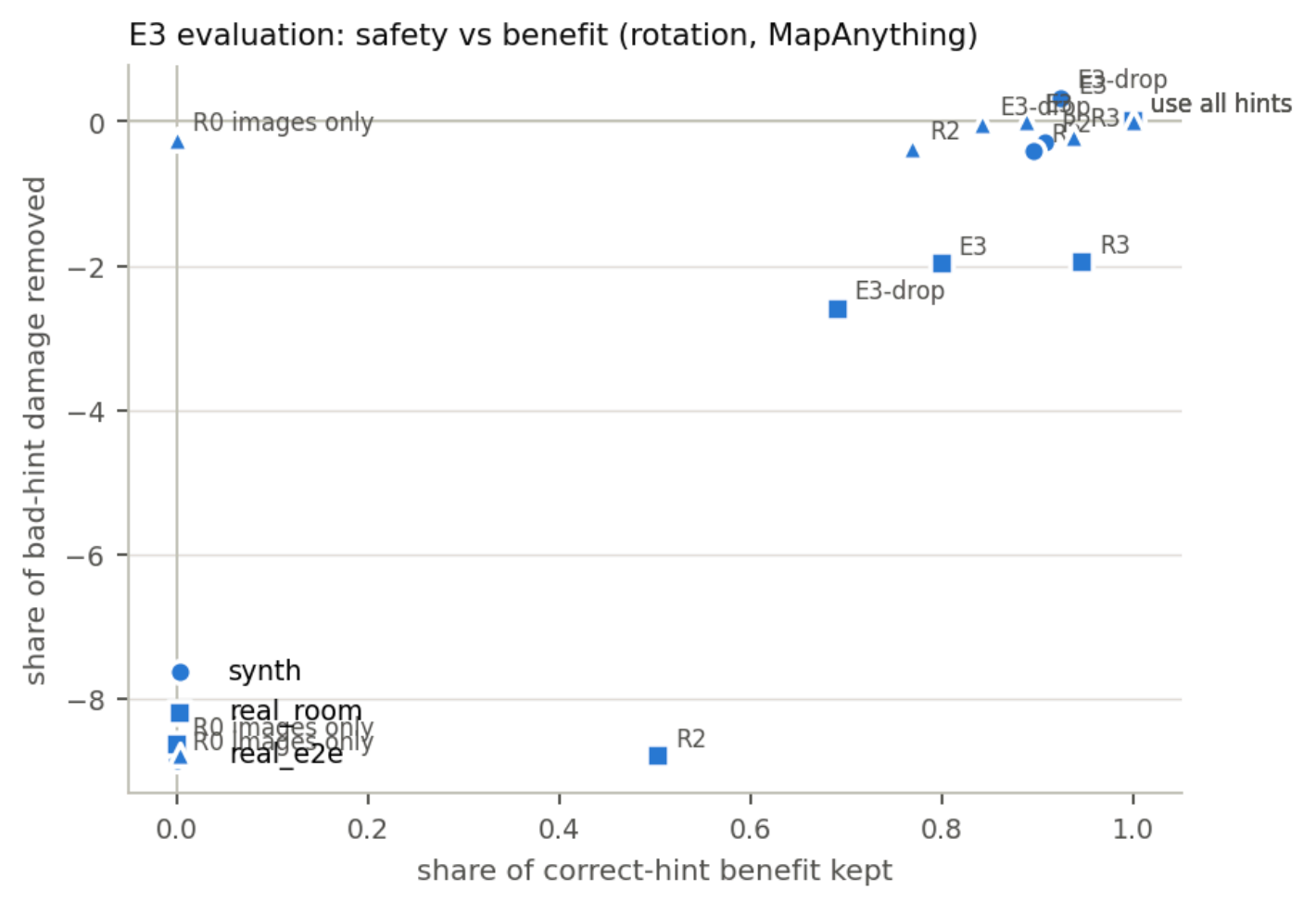
data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	harmful
synth	clean_pose	shape	hints as given	0.001	0.000	0.002	10	1.0000	False
synth	clean_pose	rotation	hints as given	0.110	0.000	0.221	10	1.0000	False
synth	clean_pose_depth2	shape	hints as given	0.001	0.000	0.003	10	1.0000	False
synth	clean_pose_depth2	rotation	hints as given	0.116	0.000	0.238	10	1.0000	False
real_room	gt_pose	shape	hints as given	0.059	0.011	0.136	15	1.0000	False
real_room	gt_pose	rotation	hints as given	4.032	0.935	7.759	15	1.0000	False
real_room	gt_pose	real_size	hints as given	0.063	0.012	0.144	15	1.0000	False
real_room	gt_all	shape	hints as given	0.070	0.017	0.153	15	1.0000	False
real_room	gt_all	rotation	hints as given	4.073	0.969	7.803	15	1.0000	False
real_room	gt_all	real_size	hints as given	0.073	0.017	0.157	15	1.0000	False
real_room	dev_pose	shape	hints as given	0.055	0.007	0.130	15	1.0000	False
real_room	dev_pose	rotation	hints as given	3.833	0.767	7.515	15	1.0000	False
real_room	dev_pose	real_size	hints as given	0.060	0.009	0.141	15	1.0000	False
real_room	dev_all	shape	hints as given	0.061	0.009	0.142	15	1.0000	False
real_room	dev_all	rotation	hints as given	3.212	0.411	6.918	15	1.0000	False
real_room	dev_all	real_size	hints as given	0.062	0.009	0.144	15	1.0000	False
real_e2e	gt_pose	shape	hints as given	0.003	0.000	0.007	15	1.0000	False
real_e2e	gt_pose	rotation	hints as given	1.015	0.000	2.887	15	1.0000	False
real_e2e	gt_pose	real_size	hints as given	0.000	-0.007	0.008	15	1.0000	False
real_e2e	gt_all	shape	hints as given	0.010	0.001	0.021	15	1.0000	False
real_e2e	gt_all	rotation	hints as given	0.984	0.002	2.775	15	1.0000	False
real_e2e	gt_all	real_size	hints as given	0.018	0.001	0.045	15	1.0000	False
real_e2e	dev_pose	shape	hints as given	0.000	-0.000	0.000	15	1.0000	False
real_e2e	dev_pose	rotation	hints as given	0.054	-0.000	0.150	15	1.0000	False
real_e2e	dev_pose	real_size	hints as given	-0.000	-0.001	0.000	15	1.0000	False
real_e2e	dev_all	shape	hints as given	0.004	-0.000	0.011	15	1.0000	False
real_e2e	dev_all	rotation	hints as given	0.052	-0.002	0.146	15	1.0000	False
real_e2e	dev_all	real_size	hints as given	0.011	-0.000	0.034	15	1.0000	False

HELP

data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	b
synth	outlier_rot_rot_deg=15	rotation	hints as given	-2.261	-2.612	-1.811	10	0.0002	T
synth	outlier_rot_rot_deg=30	rotation	hints as given	-5.261	-5.590	-4.767	10	0.0000	T
synth	outlier_rot_rot_deg=90	rotation	hints as given	-17.552	-17.640	-17.459	10	0.0000	T
synth	outlier_frac_frac=0.2	rotation	hints as given	-5.057	-5.207	-4.882	10	0.0000	T
synth	outlier_frac_frac=0.4	rotation	hints as given	-8.955	-9.357	-8.468	10	0.0000	T
synth	shared_heading_bias_rot_deg=5	rotation	hints as given	-2.308	-2.900	-1.773	10	0.0017	T
synth	shared_heading_bias_rot_deg=10	rotation	hints as given	-5.665	-6.477	-4.889	10	0.0000	T
synth	c6_shared_rot_rot_deg=5	rotation	hints as given	-2.431	-2.964	-1.958	10	0.0004	T
synth	drift_level=2	rotation	hints as given	-2.068	-2.433	-1.685	10	0.0002	T
synth	drift_level=3	rotation	hints as given	-4.546	-5.037	-4.061	10	0.0000	T
synth	depth_scale_factor=0.67	shape	hints as given	-0.032	-0.043	-0.021	10	0.0222	T
synth	depth_scale_factor=1.5	shape	hints as given	-0.042	-0.059	-0.027	10	0.0494	T
synth	depth_one_scale_factor=1.5	shape	hints as given	-0.003	-0.008	0.002	10	1.0000	F
real_room	ai_depth	shape	hints as given	-0.044	-0.101	0.015	15	1.0000	F
real_room	dev_pose_ai_depth	shape	hints as given	-0.066	-0.110	-0.000	15	1.0000	F
real_e2e	ai_depth	shape	hints as given	-0.074	-0.115	-0.041	15	0.1002	F
real_e2e	dev_pose_ai_depth	shape	hints as given	-0.061	-0.077	-0.046	15	0.0001	T

BEST

data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	bet
synth	outlier_rot_rot_deg=15	rotation	filter_rgbcheck	0.307	0.023	0.668	10	1.0000	Fals
synth	outlier_rot_rot_deg=30	rotation	filter_rgbcheck	0.228	-0.058	0.683	10	1.0000	Fals
synth	outlier_rot_rot_deg=90	rotation	filter_rgbcheck	-0.033	-0.151	0.085	10	1.0000	Fals
synth	outlier_frac_frac=0.2	rotation	filter_rgbcheck	-0.050	-0.174	0.081	10	1.0000	Fals
synth	outlier_frac_frac=0.4	rotation	filter_rgbcheck	0.604	0.255	1.012	10	0.7147	Fals
synth	shared_heading_bias_rot_deg=5	rotation	rgb_only	-1.394	-3.057	-0.488	10	1.0000	Fals
synth	shared_heading_bias_rot_deg=10	rotation	rgb_only	-1.629	-3.478	-0.582	10	1.0000	Fals
synth	c6_shared_rot_rot_deg=5	rotation	rgb_only	-1.272	-2.998	-0.240	10	1.0000	Fals
synth	drift_level=2	rotation	rgb_only	-0.865	-2.504	0.078	10	1.0000	Fals
synth	drift_level=3	rotation	rgb_only	0.037	-1.738	1.193	10	1.0000	Fals
synth	depth_scale_factor=0.67	shape	drop_lowconf	-0.031	-0.042	-0.021	10	0.0168	True
synth	depth_scale_factor=1.5	shape	drop_lowconf	-0.041	-0.059	-0.024	10	0.0826	Fals
synth	depth_one_scale_factor=1.5	shape	drop_lowconf	0.000	-0.004	0.005	10	1.0000	Fals
real_room	ai_depth	shape	rgb_only	0.009	-0.041	0.049	15	1.0000	Fals
real_room	dev_pose_ai_depth	shape	drop_lowconf	-0.059	-0.102	0.006	15	1.0000	Fals
real_e2e	ai_depth	shape	rgb_only	0.026	0.018	0.034	15	0.0011	Fals
real_e2e	dev_pose_ai_depth	shape	rgb_only	0.024	0.015	0.033	15	0.0106	Fals



data	metric	remedy	benefit_kept	damage_removed	n_bad_conditions
synth	rre_deg	none	1.000	0.000	13
synth	rre_deg	witness	0.925	0.266	13
synth	rre_deg	witness_drop	0.923	0.331	13
synth	rre_deg	filter_rgbcheck	0.907	-0.280	13
synth	rre_deg	drop_lowconf	0.896	-0.412	13
synth	rre_deg	drop_oracle	–	0.971	5
synth	rre_deg	rgb_only	0.000	-8.840	13
synth	pts_relerr_scale	none	1.000	0.000	13
synth	pts_relerr_scale	witness	0.659	0.773	13
synth	pts_relerr_scale	witness_drop	1.214	0.111	13
synth	pts_relerr_scale	filter_rgbcheck	0.823	-1.476	13
synth	pts_relerr_scale	drop_lowconf	0.938	0.179	13
synth	pts_relerr_scale	drop_oracle	–	0.939	5
synth	pts_relerr_scale	rgb_only	0.000	-2.074	13
real_room	rre_deg	none	1.000	0.000	2
real_room	rre_deg	witness	0.800	-1.966	2
real_room	rre_deg	witness_drop	0.691	-2.591	2
real_room	rre_deg	filter_rgbcheck	0.503	-8.787	1
real_room	rre_deg	drop_lowconf	0.946	-1.954	1
real_room	rre_deg	drop_oracle	–	–	0
real_room	rre_deg	rgb_only	0.000	-8.627	2
real_room	pts_relerr_scale	none	1.000	0.000	2
real_room	pts_relerr_scale	witness	0.718	0.302	2
real_room	pts_relerr_scale	witness_drop	0.709	0.365	2
real_room	pts_relerr_scale	filter_rgbcheck	0.434	-0.332	1
real_room	pts_relerr_scale	drop_lowconf	0.929	0.044	1
real_room	pts_relerr_scale	drop_oracle	–	–	0
real_room	pts_relerr_scale	rgb_only	0.000	0.073	2
real_e2e	rre_deg	none	1.000	0.000	2
real_e2e	rre_deg	witness	0.888	0.010	2
real_e2e	rre_deg	witness_drop	0.842	-0.032	2
real_e2e	rre_deg	filter_rgbcheck	0.769	-0.369	1
real_e2e	rre_deg	drop_lowconf	0.937	-0.207	1
real_e2e	rre_deg	drop_oracle	–	–	0
real_e2e	rre_deg	rgb_only	0.000	-0.255	2
real_e2e	pts_relerr_scale	none	1.000	0.000	2
real_e2e	pts_relerr_scale	witness	0.959	0.647	2
real_e2e	pts_relerr_scale	witness_drop	0.864	0.774	2

data	metric	remedy	benefit_kept	damage_removed	n_bad_conditions
real_e2e	pts_relerr_scale	filter_rgbcheck	0.971	-0.036	1
real_e2e	pts_relerr_scale	drop_lowconf	0.942	0.028	1
real_e2e	pts_relerr_scale	drop_oracle	–	–	0
real_e2e	pts_relerr_scale	rgb_only	0.000	0.888	2

7. E3 method evaluation, Pi3X (experiments/E3.md, eval splits)

E3 decision (fixed before the run): method claim holds = False; SAFE = True; HELP on agreeing / AI-depth: ai_depth, c6_shared_rot_rot_deg=5, dev_pose_ai_depth, drift_level=2, drift_level=3, shared_heading_bias_rot_deg=10, shared_heading_bias_rot_deg=5; BEST on agreeing / AI-depth: none.

Caution: SAFE passes under the strict rule, but 14 SAFE tests show harm whose 95% CI excludes 0 and whose estimate exceeds the practical threshold, without surviving Holm: real_room gt_pose shape: +0.136 (CI 0.032 to 0.272); real_room gt_pose rotation: +6.051 (CI 2.286 to 10.514); real_room gt_pose real_size: +0.137 (CI 0.031 to 0.277); real_room gt_all shape: +0.146 (CI 0.040 to 0.285); real_room gt_all rotation: +6.748 (CI 2.351 to 12.257); real_room gt_all real_size: +0.142 (CI 0.035 to 0.278); real_room dev_pose shape: +0.133 (CI 0.030 to 0.268); real_room dev_pose rotation: +5.969 (CI 2.216 to 10.362); real_room dev_pose real_size: +0.137 (CI 0.031 to 0.275); real_room dev_all shape: +0.136 (CI 0.032 to 0.269); real_room dev_all rotation: +5.654 (CI 1.950 to 10.097); real_room dev_all real_size: +0.134 (CI 0.032 to 0.265); real_e2e gt_pose rotation: +0.900 (CI 0.055 to 2.531); real_e2e gt_all rotation: +0.888 (CI 0.039 to 2.536). E3 is therefore not shown to be safe on these data.

SAFE

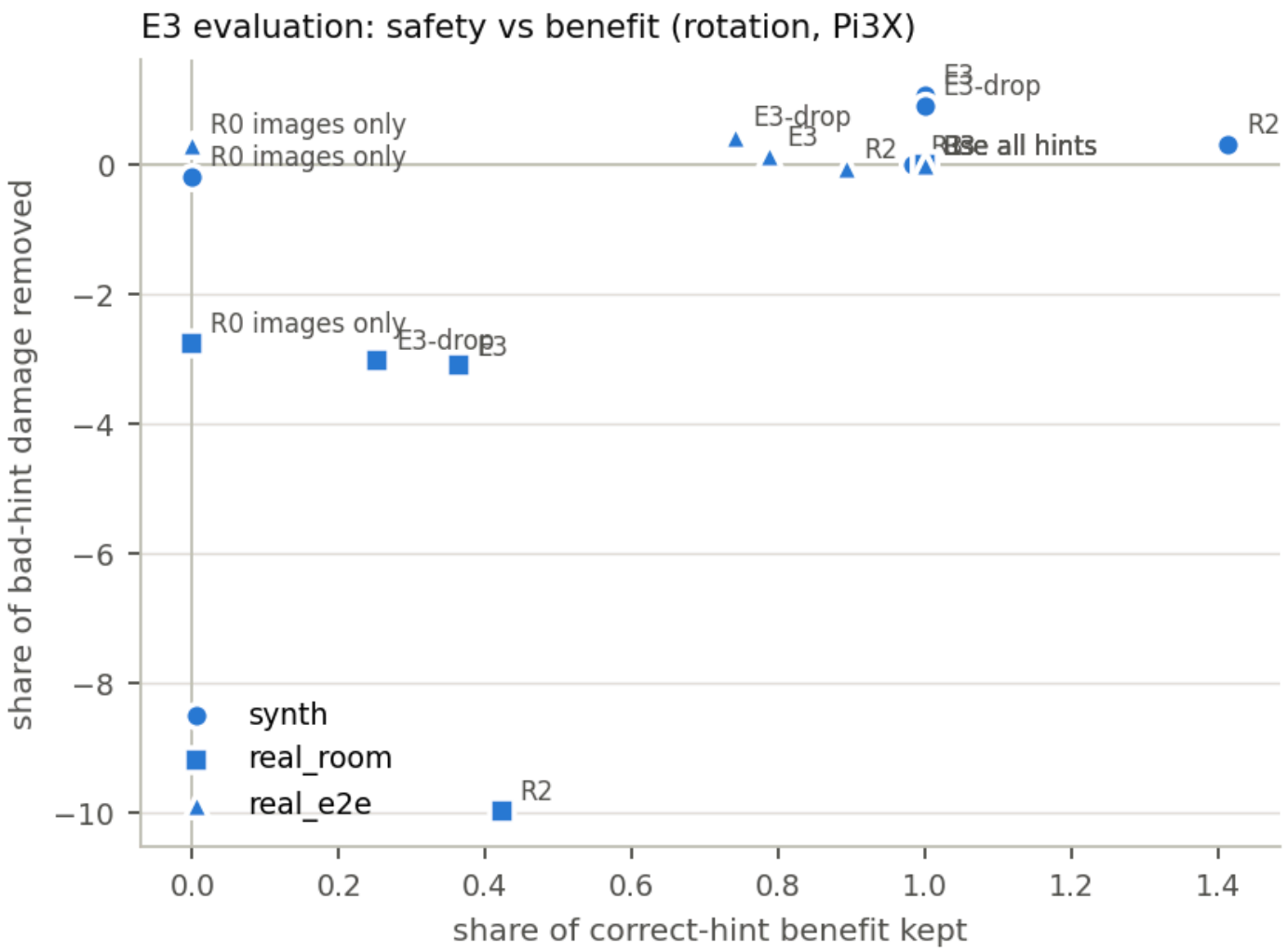
data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	harmful
synth	clean_pose	shape	hints as given	0.000	0.000	0.000	10	1.0000	False
synth	clean_pose	rotation	hints as given	0.000	0.000	0.000	10	1.0000	False
synth	clean_pose_depth2	shape	hints as given	0.000	0.000	0.000	10	1.0000	False
synth	clean_pose_depth2	rotation	hints as given	0.000	0.000	0.000	10	1.0000	False
real_room	gt_pose	shape	hints as given	0.136	0.032	0.272	15	1.0000	False
real_room	gt_pose	rotation	hints as given	6.051	2.286	10.514	15	0.6422	False
real_room	gt_pose	real_size	hints as given	0.137	0.031	0.277	15	1.0000	False
real_room	gt_all	shape	hints as given	0.146	0.040	0.285	15	1.0000	False
real_room	gt_all	rotation	hints as given	6.748	2.351	12.257	15	0.9015	False
real_room	gt_all	real_size	hints as given	0.142	0.035	0.278	15	1.0000	False
real_room	dev_pose	shape	hints as given	0.133	0.030	0.268	15	1.0000	False
real_room	dev_pose	rotation	hints as given	5.969	2.216	10.362	15	0.6422	False
real_room	dev_pose	real_size	hints as given	0.137	0.031	0.275	15	1.0000	False
real_room	dev_all	shape	hints as given	0.136	0.032	0.269	15	1.0000	False
real_room	dev_all	rotation	hints as given	5.654	1.950	10.097	15	0.8205	False
real_room	dev_all	real_size	hints as given	0.134	0.032	0.265	15	1.0000	False
real_e2e	gt_pose	shape	hints as given	0.004	0.001	0.010	15	1.0000	False
real_e2e	gt_pose	rotation	hints as given	0.900	0.055	2.531	15	1.0000	False
real_e2e	gt_pose	real_size	hints as given	0.004	0.001	0.009	15	1.0000	False
real_e2e	gt_all	shape	hints as given	0.005	0.001	0.012	15	1.0000	False
real_e2e	gt_all	rotation	hints as given	0.888	0.039	2.536	15	1.0000	False
real_e2e	gt_all	real_size	hints as given	0.005	0.001	0.010	15	1.0000	False
real_e2e	dev_pose	shape	hints as given	0.001	0.000	0.002	15	0.9015	False
real_e2e	dev_pose	rotation	hints as given	0.047	0.009	0.088	15	1.0000	False
real_e2e	dev_pose	real_size	hints as given	0.001	0.000	0.002	15	0.9015	False
real_e2e	dev_all	shape	hints as given	0.001	-0.000	0.002	15	1.0000	False
real_e2e	dev_all	rotation	hints as given	0.024	0.001	0.051	15	1.0000	False
real_e2e	dev_all	real_size	hints as given	0.001	-0.000	0.001	15	1.0000	False

HELP

data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	bet
synth	outlier_rot_rot_deg=15	rotation	hints as given	-0.297	-0.370	-0.225	10	0.0019	Tru
synth	outlier_rot_rot_deg=30	rotation	hints as given	-0.968	-1.322	-0.665	10	0.0215	Tru
synth	outlier_rot_rot_deg=90	rotation	hints as given	-9.208	-11.874	-6.313	10	0.0092	Tru
synth	outlier_frac_frac=0.2	rotation	hints as given	-0.566	-0.674	-0.457	10	0.0003	Tru
synth	outlier_frac_frac=0.4	rotation	hints as given	-1.305	-1.491	-1.094	10	0.0000	Tru
synth	shared_heading_bias_rot_deg=5	rotation	hints as given	-0.293	-0.381	-0.219	10	0.0051	Tru
synth	shared_heading_bias_rot_deg=10	rotation	hints as given	-0.762	-0.987	-0.565	10	0.0051	Tru
synth	c6_shared_rot_rot_deg=5	rotation	hints as given	-0.268	-0.384	-0.172	10	0.0497	Tru
synth	drift_level=2	rotation	hints as given	-0.498	-0.646	-0.350	10	0.0079	Tru
synth	drift_level=3	rotation	hints as given	-1.431	-1.822	-1.056	10	0.0040	Tru
synth	depth_scale_factor=0.67	shape	hints as given	-0.009	-0.016	-0.003	10	1.0000	Fals
synth	depth_scale_factor=1.5	shape	hints as given	-0.005	-0.007	-0.002	10	0.6422	Fals
synth	depth_one_scale_factor=1.5	shape	hints as given	-0.051	-0.060	-0.041	10	0.0003	Tru
real_room	ai_depth	shape	hints as given	-0.192	-0.320	-0.085	15	0.3744	Fals
real_room	dev_pose_ai_depth	shape	hints as given	0.000	-0.096	0.124	15	1.0000	Fals
real_e2e	ai_depth	shape	hints as given	-0.083	-0.115	-0.056	15	0.0055	Tru
real_e2e	dev_pose_ai_depth	shape	hints as given	-0.083	-0.105	-0.063	15	0.0002	Tru

BEST

data	condition	metric	vs	delta	ci_lo	ci_hi	n_scenes	p_holm	bet
synth	outlier_rot_rot_deg=15	rotation	filter_rgbcheck	-0.015	-0.054	0.022	10	1.0000	Fals
synth	outlier_rot_rot_deg=30	rotation	filter_rgbcheck	-0.017	-0.042	0.004	10	1.0000	Fals
synth	outlier_rot_rot_deg=90	rotation	filter_rgbcheck	-0.012	-0.046	0.019	10	1.0000	Fals
synth	outlier_frac_frac=0.2	rotation	filter_rgbcheck	-0.008	-0.019	0.001	10	1.0000	Fals
synth	outlier_frac_frac=0.4	rotation	filter_rgbcheck	0.044	-0.018	0.130	10	1.0000	Fals
synth	shared_heading_bias_rot_deg=5	rotation	rgb_only	-0.165	-0.260	-0.076	10	0.4244	Fals
synth	shared_heading_bias_rot_deg=10	rotation	rgb_only	-0.165	-0.260	-0.076	10	0.4244	Fals
synth	c6_shared_rot_rot_deg=5	rotation	rgb_only	-0.165	-0.260	-0.076	10	0.4244	Fals
synth	drift_level=2	rotation	rgb_only	-0.082	-0.194	0.026	10	1.0000	Fals
synth	drift_level=3	rotation	rgb_only	0.010	-0.120	0.155	10	1.0000	Fals
synth	depth_scale_factor=0.67	shape	drop_lowconf	-0.009	-0.016	-0.003	10	1.0000	Fals
synth	depth_scale_factor=1.5	shape	drop_lowconf	-0.004	-0.007	-0.002	10	0.6347	Fals
synth	depth_one_scale_factor=1.5	shape	drop_lowconf	-0.051	-0.061	-0.041	10	0.0003	True
real_room	ai_depth	shape	rgb_only	0.008	-0.038	0.042	15	1.0000	Fals
real_room	dev_pose_ai_depth	shape	drop_lowconf	0.000	-0.096	0.124	15	1.0000	Fals
real_e2e	ai_depth	shape	rgb_only	0.031	0.022	0.040	15	0.0006	Fals
real_e2e	dev_pose_ai_depth	shape	rgb_only	0.023	0.014	0.032	15	0.0076	Fals


E3 evaluation: safety vs benefit (rotation, Pi3X)
share of bad-hint damage removed
share of correct-hint benefit kept
0
−2
−4
−6
−8
−10
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
R0 images only
R0 images only
R0 images only
E3-drop
E3
E3-drop
E3
R2
R2
use all hints
R2
R2
synth
real_room
real_e2e

data	metric	remedy	benefit_kept	damage_removed	n_bad_conditions
synth	rre_deg	none	1.000	0.000	13
synth	rre_deg	witness	1.000	1.083	13
synth	rre_deg	witness_drop	1.000	0.918	13
synth	rre_deg	filter_rgbcheck	1.414	0.322	13
synth	rre_deg	drop_lowconf	0.983	0.001	13
synth	rre_deg	drop_oracle	–	0.954	5
synth	rre_deg	rgb_only	0.000	-0.175	13
synth	pts_relerr_scale	none	1.000	0.000	12
synth	pts_relerr_scale	witness	1.000	1.307	12
synth	pts_relerr_scale	witness_drop	1.000	-4.962	12
synth	pts_relerr_scale	filter_rgbcheck	1.007	-8.011	12
synth	pts_relerr_scale	drop_lowconf	1.018	0.026	12
synth	pts_relerr_scale	drop_oracle	–	0.399	5
synth	pts_relerr_scale	rgb_only	0.000	-17.901	12
real_room	rre_deg	none	1.000	0.000	2
real_room	rre_deg	witness	0.364	-3.098	2
real_room	rre_deg	witness_drop	0.253	-3.012	2
real_room	rre_deg	filter_rgbcheck	0.423	-9.969	1
real_room	rre_deg	drop_lowconf	1.000	0.000	1
real_room	rre_deg	drop_oracle	–	–	0
real_room	rre_deg	rgb_only	0.000	-2.752	2
real_room	pts_relerr_scale	none	1.000	0.000	2
real_room	pts_relerr_scale	witness	0.489	0.261	2
real_room	pts_relerr_scale	witness_drop	0.409	0.213	2
real_room	pts_relerr_scale	filter_rgbcheck	0.544	-1.230	1
real_room	pts_relerr_scale	drop_lowconf	1.000	0.000	1
real_room	pts_relerr_scale	drop_oracle	–	–	0
real_room	pts_relerr_scale	rgb_only	0.000	0.273	2
real_e2e	rre_deg	none	1.000	0.000	2
real_e2e	rre_deg	witness	0.787	0.136	2
real_e2e	rre_deg	witness_drop	0.740	0.436	2
real_e2e	rre_deg	filter_rgbcheck	0.893	-0.049	1
real_e2e	rre_deg	drop_lowconf	1.000	0.000	1
real_e2e	rre_deg	drop_oracle	–	–	0
real_e2e	rre_deg	rgb_only	0.000	0.318	2
real_e2e	pts_relerr_scale	none	1.000	0.000	2
real_e2e	pts_relerr_scale	witness	0.877	0.655	2
real_e2e	pts_relerr_scale	witness_drop	0.813	0.943	2

data	metric	remedy	benefit_kept	damage_removed	n_bad_conditions
real_e2e	pts_relerr_scale	filter_rgbcheck	0.861	-0.028	1
real_e2e	pts_relerr_scale	drop_lowconf	1.000	0.000	1
real_e2e	pts_relerr_scale	drop_oracle	–	–	0
real_e2e	pts_relerr_scale	rgb_only	0.000	0.868	2

7b. POST HOC (after the preregistered verdicts; changes none of them)

The preregistered BEST test compares E3 with the best existing remedy chosen per error type, which needs knowledge of the error type. A deployable system applies one remedy to everything. Tables: each remedy applied to every ETH3D bad-pose condition (rotation, scene means). `scripts/e2x/posthoc_e3.py`.

MapAnything

remedy	n_bad_conditions	mean_bad_rot_deg	worst_bad_rot_deg	correct_hints_rot_deg
none	21	4.14	17.98	0.24
witness	21	1.14	3.65	0.35
witness_drop	21	0.88	1.80	0.35
filter_rgbcheck	21	1.84	5.33	0.38
drop_lowconf	21	3.12	10.15	0.36
rgb_only	21	2.15	2.15	2.15

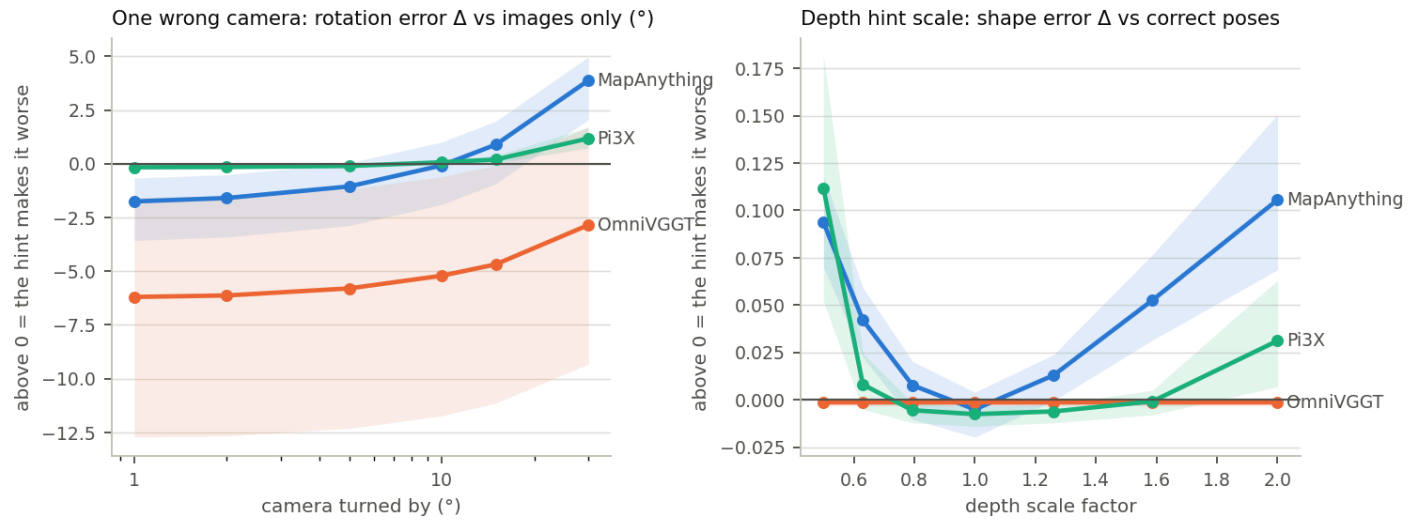
Pi3X

remedy	n_bad_conditions	mean_bad_rot_deg	worst_bad_rot_deg	correct_hints_rot_deg
none	21	1.18	9.56	0.32
witness	21	0.39	0.75	0.32
witness_drop	21	0.38	0.43	0.32
filter_rgbcheck	21	0.53	1.07	0.35
drop_lowconf	21	1.19	9.57	0.33
rgb_only	21	0.49	0.49	0.49

Where E3 acts on correct hints (calibration data only)

support	clips	acts_on_correct_hints	rate
0-2 pairs	38	13	0.342
3-9 pairs	11	2	0.182
>=10 pairs	8	0	0.000

5. Angles (DESCRIPTIVE)



Smallest error size whose 95% CI lies above 0 (the hint is then worse than going without it):

model	family	condition	size	delta_vs_floor	ci_lo	ci_hi
mapanything	c1_depth	c1_depth_factor=0.5	0.50	0.094	0.070	0.118
mapanything	c1_rot	c1_rot_rot_deg=30	30.00	3.875	2.057	4.959
pi3x	c1_depth	c1_depth_factor=0.5	0.50	0.111	0.052	0.181
pi3x	c1_rot	c1_rot_rot_deg=15	15.00	0.194	0.035	0.379
pi3x	c1_trans	c1_trans_trans_frac=3	3.00	0.284	0.137	0.420

▶ All break-even rows

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6. Geometry referee (L2, planned baseline R5; dev/cal clips)

Calibration (rule fixed in the script): view part **NOT CALIBRATED**, set part **NOT CALIBRATED**; 17% of views scored. Details: `results/e2x_geometry_referee_deg/README.md`.

ETH3D dev, one wrong camera: the referee measures the error itself.

condition	bad_view_median_deg	good_view_median_deg
outlier_rot_rot_deg=15	14.97	1.01
outlier_rot_rot_deg=30	29.86	0.95
outlier_rot_rot_deg=5	5.17	0.82
outlier_rot_rot_deg=90	89.96	1.01

---

8. Files

- `results/e2x_agg/angle_break_even.csv`
- `results/e2x_agg/angle_safety_preview.csv`
- `results/e2x_agg/xa_behaviour_ma_vs_pi3x.csv`

- results/e2x_agg/xa_behaviour_pi3x.csv
- results/e2x_agg/xa_family_ma_vs_pi3x.csv
- results/e2x_agg/xa_harm_pi3x.csv
- results/e2x_agg/xb_e2e_tests_pi3x.csv
- results/e2x_agg/xb_room_real_errors.csv
- results/e2x_agg/xb_room_remedies_mapanything.csv
- results/e2x_agg/xb_room_remedies_omnivggt.csv
- results/e2x_agg/xb_room_remedies_pi3x.csv
- results/e2x_agg/xb_room_tests_mapanything.csv
- results/e2x_agg/xb_room_tests_omnivggt.csv
- results/e2x_agg/xb_room_tests_pi3x.csv
- results/e2x_e2e_px_dev/per_run.csv
- results/e2x_e2e_px_eval/per_run.csv
- results/e2x_e3cal_agg/mapanything/chosen.json
- results/e2x_e3cal_agg/mapanything/detection_rates.csv
- results/e2x_e3cal_agg/mapanything/false_action_rates.csv
- results/e2x_e3cal_agg/mapanything_secondary/chosen.json
- results/e2x_e3cal_agg/mapanything_secondary/false_action_rates.csv
- results/e2x_e3cal_agg/pi3x/chosen.json
- results/e2x_e3cal_agg/pi3x/detection_rates.csv
- results/e2x_e3cal_agg/pi3x/false_action_rates.csv
- results/e2x_e3cal_agg/pi3x_secondary/chosen.json
- results/e2x_e3cal_agg/pi3x_secondary/false_action_rates.csv
- results/e2x_e3cal_real_ma_cal/per_run.csv
- results/e2x_e3cal_real_ma_roomdev/per_run.csv
- results/e2x_e3cal_real_px_cal/per_run.csv
- results/e2x_e3cal_synth_ma_cal/per_run.csv
- results/e2x_e3cal_synth_ma_eth3d/per_run.csv
- results/e2x_e3cal_synth_ma_tum/per_run.csv
- results/e2x_e3cal_synth_ov_eth3d/per_run.csv
- results/e2x_e3cal_synth_px_cal/per_run.csv
- results/e2x_e3cal_synth_px_eth3d/per_run.csv
- results/e2x_e3cal_synth_px_tum/per_run.csv
- results/e2x_geometry_referee_deg/chosen.json
- results/e2x_geometry_referee_deg/eth3d_dev_descriptive.csv
- results/e2x_pi3x_checks/per_run.csv
- results/e2x_pi3x_eth3d_dev/per_run.csv
- results/e2x_pi3x_eth3d_eval/per_run.csv
- results/e2x_room_ma_dev/per_run.csv
- results/e2x_room_ma_eval/per_run.csv
- results/e2x_room_ov_dev/per_run.csv
- results/e2x_room_ov_eval/per_run.csv
- results/e2x_room_px_dev/per_run.csv
- results/e2x_room_px_eval/per_run.csv
- results/e2x_xm_agg/m1_curve.csv
- results/e2x_xm_agg/m1_dial.csv

- results/e2x_xm_agg/m2_m4_state.csv
- results/e2x_xm_agg/m3_patch.csv
- results/e2x_xm_agg/m3_point_of_no_return.csv
- results/e2x_xm_agg/m5_spread.csv
- results/e2x_xm_agg/m8_scale.csv
- results/e3_eval_agg/e3_safety_benefit.csv
- results/e3_eval_agg/e3_tests.csv
- results/e3_eval_agg/e3_verdict.json
- results/e3_eval_agg/posthoc_fixed_remedies.csv
- results/e3_eval_agg/posthoc_support_vs_false_actions.csv
- results/e3_eval_agg_pi3x/e3_safety_benefit.csv
- results/e3_eval_agg_pi3x/e3_tests.csv
- results/e3_eval_agg_pi3x/e3_verdict.json
- results/e3_eval_agg_pi3x/posthoc_fixed_remedies.csv
- results/vm_logs_e2x/machine.txt
- results/vm_logs_e2x/mono_verify.txt
- results/vm_logs_e2x/session.txt